

$$\dot{\hat{x}} = A\hat{x} + c + K(z - H\hat{x})$$

$K(t)$ ojačevalni faktorji $\rightarrow$ jednako K_{∞} namesto $K(t)$

Senzor n-toga reda

$$\frac{d^n x}{dt^n} = 0 + w$$

Senzor 1. reda



Je optimalen za sledenje

$$\begin{aligned} \dot{x} &= 0 + w \\ \langle w(t) | w(t') \rangle &= Q \delta(t - t') \\ z &= x + r \\ \langle r | r \rangle &= R \end{aligned}$$

$$\begin{aligned} S: \quad & \dot{\hat{x}} = 0 + w \\ & \begin{bmatrix} A=0 \\ C=0 \\ H=1 \\ \Gamma=1 \end{bmatrix} \rightarrow \begin{bmatrix} \dot{\hat{x}} = 0 + w \\ \hat{p} = Q - P^2 R \\ K_{\infty} = \frac{P_{\infty}}{R} = \sqrt{\frac{Q}{R}} \\ \dot{\hat{x}} = \sqrt{\frac{Q}{R}} (z - \hat{x}) \\ \tau \dot{\hat{x}} + \hat{x} = z \quad \tau = 1/K_{\infty} \end{bmatrix} \end{aligned}$$

Npr: termometer

$$p = -\frac{\lambda S}{d} (T_2 - T)$$

$$p = \frac{dQ}{dt} = m c_p \frac{dT}{dt}$$

$$\dot{T} = \frac{\lambda S}{m c_p} (T_2 - T)$$



$$\frac{1}{\lambda} \dot{T} + T = T_2$$

Tipični vhodi

- 1) $z(t) = \delta(t)$
- 2) $z(t) = H(t)$
- 3) $z(t) = t^n$
- 4) $z(t) = \cos \omega t$

1. red

$$z(t) = \delta(t) \rightarrow \boxed{u=1} \rightarrow G(t) = ?$$

$$\tau \dot{x} + x = \delta(t)$$

1) homogeni del

$$\dot{x} = -\frac{1}{\tau} x \rightarrow x = C e^{-t/\tau}$$

2) part. reš.

$$\tau \dot{x} + x = \delta(t) \int_{-\infty}^t dt, \lim_{t \rightarrow 0} \rightarrow$$

$$\tau (x(t) - x(t/\tau)) + 0 = 1$$

$$x(0)$$

$$\tau x(0) = 1 \rightarrow x(0) = \frac{1}{\tau}$$

$$\rightarrow \boxed{G(t) = \frac{1}{\tau} e^{-t/\tau}}$$

$$z(t) = \alpha t$$

1) hom. del

$$x(t) = C e^{-t/\tau}$$

2) part. del

$$\text{postavka } x = At + B$$

$$\tau A + At + B = \alpha t$$

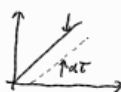
$$A = \alpha, B = -\alpha \tau$$

$$\rightarrow x(t) = C e^{-t/\tau} + \alpha t - \alpha \tau$$

$$\text{RD } x(0) = 0$$

$$C = \alpha \tau$$

$$x(t) = \alpha \tau e^{-t/\tau} + \alpha(t - \tau)$$



Senzor 2. reda

optimalen za dinamiko $\ddot{x} = 0 + w$

$$\dot{x} = v \rightarrow \dot{\hat{x}} = \begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix}$$

$$\dot{\hat{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \hat{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w$$



$$H = [1 \ 0]$$

$$M: \hat{\hat{x}} = \begin{bmatrix} \hat{x} \\ \hat{v} \end{bmatrix}$$

$$\text{Kalman: } \dot{\hat{\hat{x}}} = A\hat{\hat{x}} + PH^T R^{-1} (z - H\hat{\hat{x}})$$

$$\dot{P} = AP + PA^T + \Gamma Q \Gamma^T - PH^T R^{-1} HP \quad R = \langle v v^T \rangle \text{ skalar}$$

$$= \begin{bmatrix} 2P_{12} & P_{22} \\ P_{12} & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & Q \end{bmatrix} - \frac{1}{R} \begin{bmatrix} P_{11} & P_{12} & P_{12} \\ P_{12} & P_{22} & P_{22} \end{bmatrix}$$

$$\dot{P}_{11} = 2P_{12} - \frac{1}{R} P_{11}^2$$

$$\dot{P}_{12} = P_{22} - \frac{1}{R} P_{11} P_{12}$$

$$\dot{P}_{22} = Q - \frac{1}{R} P_{12}^2$$

$$\dot{P}_{11} = 2P_{12} - \frac{1}{R} P_{11}^2$$

$$\dot{P}_{12} = P_{22} - \frac{1}{R} P_{11} P_{12}$$

$$\dot{P}_{22} = Q - \frac{1}{R} P_{12}^2$$

stac. reš.

$$P_{11} = \sqrt{QR}$$

$$P_{12} = \sqrt{2R P_{11}} = \sqrt{2R} \sqrt{\sqrt{QR}}$$

$$P_{22} = \frac{1}{R} \sqrt{2R Q^2} \sqrt{\sqrt{QR}}$$

Ojačevalni faktorji

$$K_{\infty} = PH^T R^{-1} = \frac{1}{R} \begin{bmatrix} P_{11} \\ P_{12} \end{bmatrix}$$

$$\dot{\hat{x}} = \hat{v} + \frac{1}{R} P_{11} (z - \hat{x}) \quad / \frac{d}{dt}$$

$$\dot{\hat{v}} = 0 + \frac{1}{R} P_{12} (z - \hat{x})$$

$$\ddot{\hat{x}} = \frac{1}{R} P_{12} (z - \hat{x}) + \frac{1}{R} P_{11} (\dot{z} - \dot{\hat{x}})$$

$$\ddot{\hat{x}} + \frac{P_{11}}{R} \dot{\hat{x}} + \frac{P_{12}}{R} \hat{x} = \frac{P_{11}}{R} \dot{z} + \frac{P_{12}}{R} \frac{1}{z}$$

standardna notacija

$$\ddot{\hat{x}} + 2\zeta\omega\dot{\hat{x}} + \omega^2\hat{x} = 2\zeta\omega\dot{z} + \omega^2 z$$

$$\omega^2 = \frac{P_{11}}{R} = \sqrt{\frac{Q}{R}}$$

$\rightarrow$ za večino senzorja 2. reda $= 0$

$$2\zeta\omega = \frac{P_{12}}{R} = \frac{1}{R} \sqrt{2R P_{11}}$$

$$\zeta = \frac{1}{\sqrt{2}}$$

Rešiti sekvencijalno 2. reda za tipične signale

1) $z(t) = \delta(t)$, $x(t) = G(t) = ?$

$$\ddot{x} + 2\xi\omega\dot{x} + \omega^2x = \omega^2\delta(t) \quad \int_{-\infty}^t dt, \lim_{t \rightarrow 0}$$

$$\dot{x}(t) - \dot{x}(t^-) + 2\xi\omega(x(t) - x(t^-)) + 0 = \omega^2$$

$$\dot{x}(0) + 2\xi\omega x(0) = \omega^2$$

$$x(0) = 0$$

$$\dot{x}(0) = \omega^2$$

Hom. del

$$\ddot{x} + 2\xi\omega\dot{x} + \omega^2x = 0$$

$$\lambda^2 + 2\xi\omega\lambda + \omega^2 = 0$$

$$\lambda_{1,2} = -\frac{2\xi\omega \pm \sqrt{4\xi^2\omega^2 - 4\omega^2}}{2} = -\omega\left(\xi \pm \sqrt{\xi^2 - 1}\right) = -\omega\left(\xi \pm i\sqrt{1 - \xi^2}\right) = -\omega\left(\frac{1}{\sqrt{2}} \pm i\frac{1}{\sqrt{2}}\right) = -\frac{\omega}{\sqrt{2}}(1 \mp i)$$

Nastavak:

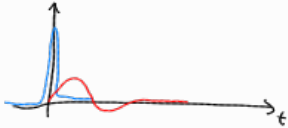
$$x = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

$$x(0) = 0 \Rightarrow C_1 = -C_2$$

$$\dot{x}(0) = 0 \Rightarrow C_1 \lambda_1 + C_2 \lambda_2 = \omega^2$$

$$C_1 = \frac{\omega^2}{\lambda_1 - \lambda_2} = \frac{\omega^2}{-\frac{\omega}{\sqrt{2}}(-2i)} = \frac{\omega}{\sqrt{2}i} = -\frac{i\omega}{\sqrt{2}}$$

$$\rightarrow x = -\frac{i\omega}{\sqrt{2}} e^{-\frac{\omega}{\sqrt{2}}(1-i)t} + \frac{i\omega}{\sqrt{2}} e^{-\frac{\omega}{\sqrt{2}}(1+i)t} = \frac{\omega\sqrt{2}}{2i} e^{-\frac{\omega}{\sqrt{2}}t} 2i \sin \frac{\omega t}{\sqrt{2}} = \omega\sqrt{2} \sin \frac{\omega t}{\sqrt{2}} e^{-\frac{\omega t}{\sqrt{2}}}$$



Splošna ξ :

$$\lambda_{1,2} = -\xi\omega \pm i\omega\sqrt{1-\xi^2}$$

$$G(t) = \frac{\omega}{\sqrt{1-\xi^2}} \sin \sqrt{1-\xi^2} \omega t e^{-\xi\omega t}$$

$$\xi \rightarrow 0: x(t) = \omega \sin \omega t$$

$$\xi \gg 1: x(t) = \frac{\omega}{i\sqrt{\xi^2-1}} \left[\frac{e^{-\omega\sqrt{\xi^2-1}t} - e^{-\omega\sqrt{\xi^2+1}t}}{2i} \right] e^{-\xi\omega t} =$$

$$= \frac{\omega}{\sqrt{\xi^2-1}} \sinh(\sqrt{\xi^2-1}\omega t) e^{-\xi\omega t} \approx \omega^2 t e^{-\xi\omega t}$$